Optimization of Cross Correlation Algorithm for Annual Mapping of Alpine Glacier Flow Velocities; Application to Sentinel-2

Jérémie Mouginot, Antoine Rabatel, Etienne Ducasse, and Romain Millan

***Abstract*— Nowadays, satellite observations cover most of the Earth's surface in a repetitive manner. This information is crucial for documenting variability and environmental changes such as glacier surface velocity. With this in mind, digital image processing has been developed and improved over the past decades. The processing challenges are now related to optimizing parameters that account for the high variability of natural processes, as well as filtering and aggregating the results to provide useful products to end-users. Based on the normalized cross correlation (NCC) method applied to Sentinel-2 optical satellite observations up to 400 days apart, we present a series of tests to derive optimal parameter values for the quantification of alpine glacier ice velocity that we have applied to the Mont-Blanc massif where in situ measurements are available. We found that a search distance adapted to the temporal baseline, a 16 × 16 pixel window size, and a 5 × 5 pixels sampling provide an appropriate combination of parameters to process Sentinel-2 with the NCC method when applied to small alpine glaciers. Combining several spatial and temporal filters applied to a large set of more than 18 000 displacement maps obtained between 2015 and 2021, then aggregating these filtered maps using statistical or linear regressions into annual maps, yields near-complete maps of the test region with a root mean square error (RMSE) reduced to about 10 m.yr^{-1} compared to in situ measurements.**

***Index Terms*— Alpine glaciers, data reduction, ice velocity, normalized cross correlation (NCC), optical satellite images, sentinel-2.**

I. Introduction

GLACIERS flow downhill due to gravity, transporting ice from upper elevations where the accumulation of mass by snowfall usually dominates, to their lower reaches, where mass loss by ablation and possibly calving occurs. Ice flow is a combination of both sliding and creeping and depends on the slope of the glacier bed and the amount of nourishment. For a glacier in dynamic equilibrium, the related mass flux is balanced by surface processes, i.e., the mass transported to the terminus is the same as the mass that is lost. In such conditions, a glacier would not change its geometry over time and can be considered as "stable" (in steady state) [1]. For advancing or retreating glaciers, the dynamic equilibrium is not maintained and glacier flow velocities are higher or lower than required. Therefore, documenting the long-term evolution of glacier flow velocity reveals the past and current state of glaciers. Satellite data provide worldwide coverage of glaciers and their flow velocities but the so far applied automated and standardized processing does often not provide the spatial resolution required to resolve small alpine glaciers (e.g., smaller than 0.5–1 km^2) in steep mountain terrain, as a recent study by Millan et al. [2] has shown.

Image correlation methods are commonly used to derive glacier surface displacements from consecutive satellite observations (optical and radar) acquired over a period of time Δt (e.g., [3]). Although computationally intensive, the method is relatively simple and can measure displacement in two dimensions. The typical image matching method is based on a normalized cross correlation (NCC) algorithm (e.g., [4]), but several alternatives have been developed: cross correlation operated in the Fourier domain [5], least squares matching [6], phase correlation [7], [8], [9], orientation correlation [10], or more recently optical flow technologies applied to dense satellite images time series [11]. Heid and Kääb [12] made a thorough comparison of six image-matching algorithms commonly used in the literature using optical Landsat images. They concluded that none of the methods clearly outperforms the others in all circumstances and that the subpixel accuracy is dominated by sensor noise and illumination. However, for relatively small alpine glaciers, NCC outperforms all other tested methods. Heid and Kääb [12] encouraged that future developments be made not in the matching algorithms themselves, but rather in the way they are implemented by testing different parameterizations of one algorithm and comparing the results to select an optimal result. Although sensor noise and illumination differ in Landsat and Sentinel-2 data, we assume here that the results would be similar for Sentinel-2. Indeed, the specifications of the spectral bands used for the correlation are very close between thematic mapper (TM), enhanced TM Plus (ETM+), and operational land imager (OLI) sensors onboard Landsat missions and multispectral instrument (MSI) onboard Sentinel-2. Furthermore, one of the conclusions of the European Space Agency (ESA) Glaciers Climate Change Initiative project [13] was to develop algorithms and processing chains that allow a high degree of automation in order to be able to process massive amounts of

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satellite data over large areas. Building on these earlier studies, we focus here only on the NCC method applied to Sentinel-2 optical satellite data. Three main parameters can be adjusted and directly influence the obtained results: the window size of the subimages that are correlated, the distance that is defined to find the correlation peak and the sampling resolution.

To date, no time series of alpine glacier flow has been established. Due to the difficulty of measuring the flow in winter with optical sensors and the high noise associated with the measurements, it seems appropriate to consider the multiyear evolution of the flow and not their seasonal fluctuations. The goal of this work is to define the optimal way to produce automated glacier displacement fields from alpine glacier-type regions. The challenge of this mapping is to obtain accurate results despite the decametric resolution of current freely available image archives. Because of this challenge, such regions are not included in the current databases (e.g., ITS_LIVE) or with incomplete measurements and relatively unrepresentative of reality (e.g., GoLIVE, [14]). These multiannual measurements remain essential to, for example, characterize the evolution of the flow according to the changes in the geometry of the glaciers or to capture potential surges. These results could allow a better understanding of the properties and physical laws that govern the flow of ice [15].

The objective of this study is therefore to analyze how the accuracy of the NCC matching algorithm, but also how the post-processing of a large and relatively noisy dataset could be improved for annual velocities of small Alpine glaciers using Sentinel-2 data, as these glaciers have not received enough attention in other studies. In the following, we first test different NCC parameters to derive optimal values for quantifying surface displacement maps of Alpine glaciers from the use of satellite images recorded with a Δt ranging from 5 to 400 days. We then investigate different data filtering and aggregation approaches to remove outliers and propose complete annual maps of alpine glacier surface flow velocity.

II. Data and Methods

A. Description of Sentinel-2 and in situ Observations

Table I summarizes the main characteristics and limitations of the data acquired by Sentinel-2A and 2B (S2), which are optical imagers developed and operated by the European Space Agency in the framework of the Copernicus program. In the current study, we used all available images recorded by Sentinel-2 over the Mont-Blanc massif located in the French Alps (at the border with Italy and Switzerland) corresponding to the tile number T32TLR.

For validation of ice velocities derived with S2, we use in situ measurements from differential navigation satellite systems (d-GNSS). These data consist of topographical measurements made at the location of ablation stakes on the glacier surface and have an annual temporal resolution and decimeter precision. In the Mont-Blanc massif, such data are available on several glaciers through the Service National d'Observation les GLACIers, un Observatoire du CLIMat (GLACIOCLIM) (https://glacioclim.osug.fr/) since 1976 (Fig. 1). It is important to note that the d-GNSS measurements are point measurements while the velocity maps obtained have a resolution of 50 × 50 m. Thus the comparison of the two is not perfect and, even if we believe that the differences are small enough to evaluate the different filtering and aggregation methods, it is possible that biases are still present. Furthermore, we assume that the d-GNSS data represent the ground truth, while it is possible that they are subject to small error or bias.

TABLE I

Characteristics of the Data Used for the Production of Glacier Surface Flow Velocity Maps

	Sentinel-2
Spectral resolution	13 band (0.4-2.2 μm)
Spatial resolution	10-60 m (depending on the band)
Temporal resolution	5-10 days
Time frame	S2A since mid-2015 and S2B since spring 2017
Incidence angle	Nadir
Costs	Free
Cloud problems	Potentially high
Availability	available as an ortho-rectified product

B. Description of the Cross Correlation

Feature tracking approaches measure the displacement of features between two consecutive images. In the 1990s, the development of feature tracking algorithms based on image matching methods allowed the automation of image processing and increased the efficiency and accuracy of the method [3], [4]. Image matching relies on finding a similar pattern from one image to another at a different time period. The similarity measure is classically implemented through NCC.

The core of the processing to retrieve surface displacements is the correlation stage. As mentioned in the Introduction, Heid and Kääb [12] indicated that performances of existing cross correlation methods are quite similar and that the NCC method is the most suited for small mountain glaciers like in the European Alps. Therefore, we propose to test several sets of values for each of the three parameters used in the NCC method. These parameters are: 1) the search distance; 2) the window size; and 3) the sampling resolution. Indeed, the NCC parameters used for the ice sheets or other large glacierized areas, might not be optimal for the alpine-type glaciers that are relatively small in size compared to the native resolution of the sensor considered here (10 m in the case of S2). Therefore, we investigate which set of NCC parameter values is most appropriate for alpine-type glaciers.

The NCC coefficient can be written as

$$\mathrm{NCC}(u,v)=\frac{\sum_{x=0}^{n_x}\sum_{y=0}^{n_y}\left[R(x,y)-\bar{R}\right]\left[S(x-u,y-v)-\bar{S}\right]}{\sqrt{\sum_{x=0}^{n_x}\sum_{y=0}^{n_y}\left(R(x,y)-\bar{R}\right)^2}\sqrt{\sum_{x=0}^{n_x}\sum_{y=0}^{n_y}\left(S(x,y)-\bar{S}\right)^2}} \tag{1}$$

where R is the reference image and S is the secondary image, $n_x \times n_y$ is the window size of the subimages that are correlated, u and v are the displacement in x- and y-direction between R and S, respectively. The maximum values u and v correspond to the search distance provided by the algorithm. The NCC is generally not performed for every pixel of the input images R and S, so the sampling resolution represents the step at which the NCC evaluation is performed.

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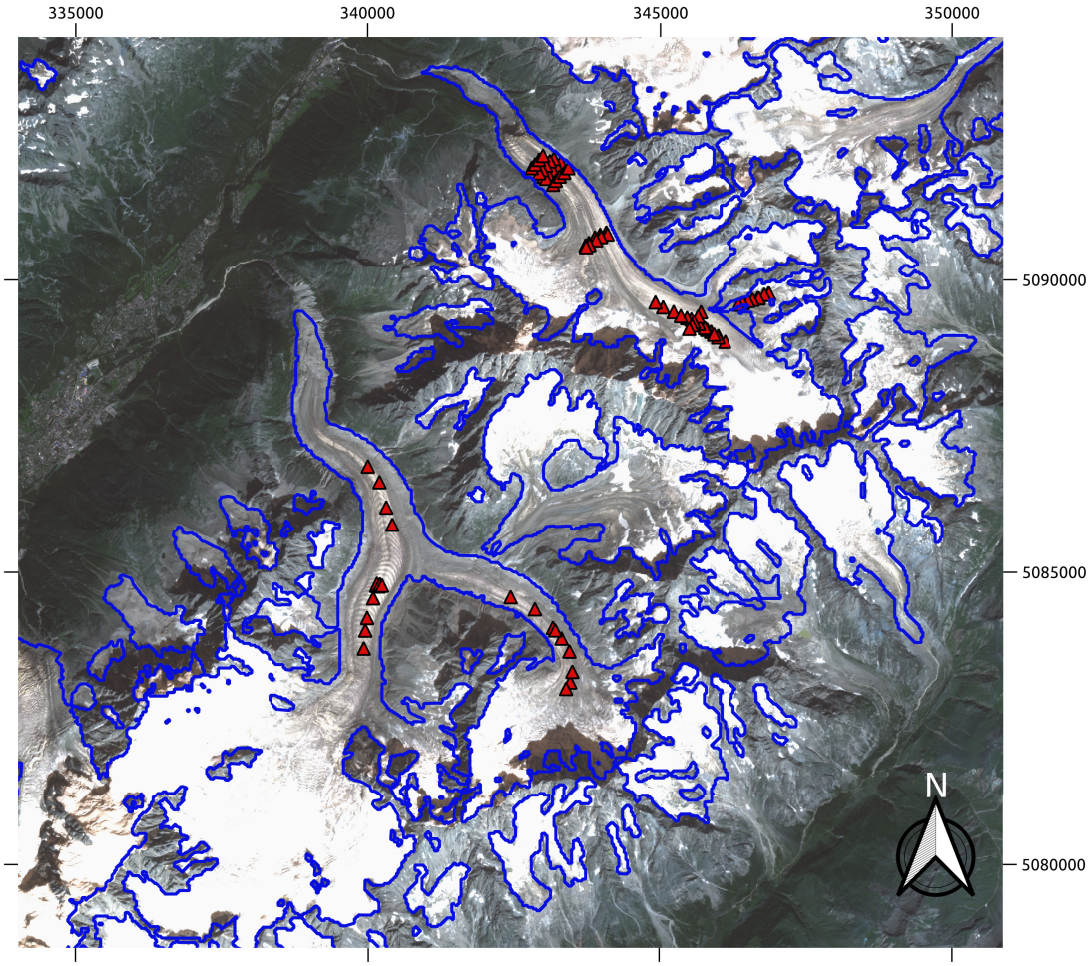


Fig. 1. Study area in the Mont-Blanc massif. Blue outlines show the glaciers extent taken from the Randolph Glacier Inventory [16]. Red triangles show the location of the glacier ablation stakes used for d-GNSS measurements of glacier surface flow velocity. The background is a Sentinel-2 color composite image acquired on 2018/08/28. *Argentière Glacier* is the one located to the North-East (where stakes can be seen), the other being the *Mer de Glace*.

C. Description of the Work-Flow

Using the optimized NCC parameters, we process the S2 image archive available since 2015 with less than 60% cloud cover over the Mont-Blanc massif following the workflow described in [17]. As recommended by Kääb et al. [18], band 8 is used. This band in the near-infrared spectra seems to provide good correlation results over glacierized area [18], but a combination of different bands or another one might also prove useful in certain cases (not tested here). As a summary, our work-flow for S2 includes: 1) a preprocessing of the images (application of a Sobel filter in the x-and y-direction to enhanced surface features [19]); 2) the quantification of the displacements with the NCC method based on a modified version of `ampcor` [17]; and 3) displacement maps are calibrated on the stable areas around the glaciers to correct potential biases due to geometric distortions or geolocation errors.

We modified the Jet Propulsion Laboratory (JPL)-developed `ampcor` algorithm that was part of the repeated orbit interferometry package (ROI-PAC) [20]. We use the current cross correlation algorithm, written in Fortran-90, but it has been integrated in a Python environment allowing some flexibility in the management of the correlation. The Python environment extracts the chips from the master and slave images and calls `ampcor` using the `numpy.f2py` interface. The Fortran-90 code of `ampcor` computes a NCC map between the reference image chip and the slave chips. From this correlation map, a maximum correlation value is determined. The reference and slave image chips are deramped and oversampled by a factor of two using a 2-D fast Fourier transform. The correlation procedure is then repeated around the previously determined peak value. The resulting new correlation map is itself oversampled by a factor of two by Fourier transform and the location of the correlation peak is then determined. The performance of the correlation called signal-to-noise ratio (SNR) is computed as the squared ratio between the correlation maximum and the average of the surrounding correlation values. The displacement maps in pixels are then converted to glacier surface velocity in meters per year. The x- and y-component of the displacements in meters per year are stored as GeoTiff at the sampling resolution. We use the Geospatial Data Abstraction Library (GDAL) for all geographical transformations and formatting of final files. To facilitate post-processing, the maps are divided into tiles of equal size of 12.5 $\times$ 12.5 km with a resolution of 50 m (or 250 $\times$ 250 pixels) as described in [17]. For each tile, we extract and

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stack the velocity maps and associated metadata into "cubes" whose time dimension therefore corresponds to the number of calculated speed maps. All repeat cycles (same orbit) from 5 to 400 days are processed in our region of interest and stored following this approach.

D. Data Reduction Approaches

Once glacier surface velocity maps are derived for different temporal baselines between 5 and 400 days, we reduce them to uniformly sampled, comprehensive maps of glacier surface flow velocities. Such data reduction consists of filtering outliers and merging/aggregating the results into a corrected, ordered, and simplified form. Here, we explore the effectiveness of different filtering and aggregation methods used to perform data reduction on the S2-derived maps of annual glacier surface flow velocity.

1) Description of the Data Filtering Approach: For filtering glacier surface flow velocity measurements, several steps are considered to eliminate outliers and improve the signal-to-noise ratio. The filtering considered here is applied to all the measurements obtained, it can be applied either to the spatial or the temporal dimension in our data cube. These filters can be used individually or in combination to improve the signal, the aim being to simplify the analysis of the results without removing the correct information. Filters considered here are.

1) Filter out obvious outliers, i.e., nonplausible values for mountain glaciers of alpine-type. In the Alps, the fastest glaciers are flowing at a pace of up to 1000 m.yr^{-1} [2], [17], [21]. Therefore, surface flow velocities faster than 1000 m.yr^{-1} can be discarded.
2) Apply a median spatial filtering. We eliminate pixels for which the pixel offsets deviate more than three units from the 9×9 median value [17], [22], [23]. This filtering is applied separately on the two components of the speed (v_x and v_y). When one component is filtered out, the other component is also removed.
3) Filter on the temporal baselines (time between two acquisitions of a pair of images used to estimate displacement) based on the glacier flow magnitude vs. the expected accuracy. Indeed, the noise associated with the measured displacements increases inversely with the temporal baseline [17]. Thus the noise associated with short cycles makes it impossible to measure the shortest displacements (see Table II). Conversely, the longest temporal baselines do not capture the fastest displacements. Based on the S2 spatial resolution and the temporal baselines between the images used to derive the displacements, we list in Table II the minimum temporal baseline that were considered for different flow velocity magnitudes. In order to test the sensitivity of the results to this filtering, we apply a multiplication factor n varying between 1 and 4 to the defined temporal baselines. The higher this factor n is, the higher the temporal baselines will be privileged.
4) Filter on flow direction as it is unexpected that the flow direction varies significantly over the study period from 2015 to 2021. We test different angles for which the direction of the velocity vector is allowed to vary around the median flow direction. Angle values from ± 5 up to $\pm 65°$ are tested. Displacement values whose direction is outside the tested range are discarded.
5) Filter on the distribution of the surface flow velocities. For each pixel, we consider the distribution of the surface flow velocities quantified over the entire study period (2015–2021). We then eliminate the distribution tails from the percentile values ranging from 2.5% to 25%. Thus, as an example, for the 2.5% and 25% percentile, 5% and 50% of the extreme values of the distribution are filtered, respectively.

TABLE II

MINIMUM VALUES OF TEMPORAL BASELINES USED FOR DIFFERENT RANGES OF GLACIER FLOW VELOCITIES. THE PARAMETER n VARIES BETWEEN 1 AND 4

Minimum temporal baselines (days)	v_{min} (m.yr^{-1})	v_{max} (m.yr^{-1})
$100\times n$	//	10
$60\times n$	10	20
$30\times n$	20	40
$20\times n$	40	60
$10\times n$	60	90
$5\times n$	90	//

The outputs of each filtering step are evaluated by comparing the results with in situ d-GNSS measurements. The d-GNSS measurements are carried out annually and therefore give a local estimate of the glacier's displacement between two dates approximately one year apart. To evaluate the results of the filtering, we calculate the median of the S2 measurements whose acquisition dates are between the d-GNSS measurement dates with a margin of ten days. The quality assessment is then done considering the root mean square error (RMSE) and the coefficient of determination (denoted r^2) between the d-GNSS and the S2 median values.

2) Description of the Data Aggregation Approaches: For the aggregation of individual velocity measurements in order to create the most accurate and complete annual glacier flow mapping in the Alps, four simple and robust aggregation methods are tested, applied individually for each pixel (every 50 m) on the filtered data.

1) Compute the median for each year from 2015 to 2021 of individual velocity measurements. Hydrological years are considered, i.e., from October 1st to September 30th of the following year. The central date between the primary and secondary acquisitions defines in which year the measurement was taken.
2) Compute the weighted average for each year from 2015 to 2021 with the same convention as for the median. The weights are defined as $1/\sigma^2$ where σ is the estimated error on the ice velocity associated with each cycle.
3) Compute the ordinary least square (OLS) (also called linear regression) regression between 2015 and 2021. From the regression, the results obtained on March first (central date for the hydrological year) would be used for the annual time series.
4) Calculate the Theil-Sen estimator which is based on the fit of a linear regression in a scatter plot by considering

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the median of the slopes of all possible lines passing through pairs of points. This linear regression is particularly robust to outliers.

Mountain glaciers are generally subject to moderate seasonal variations in their flow, which may affect the estimation of annual velocity. Theoretically, in the median and weighted average approaches, biases can occur if the sampling is not constant in time (or incomplete). Indeed, if there is a larger sampling during certain seasons of the year, the results of these two aggregation methods will tend toward the flow velocities of the most represented season in our measurement ensemble. Conversely, the OLS and Theil-Sen linear regressions take into account of the temporal aspect of sampling and are therefore less affected by incomplete or irregular sampling.

To test the robustness of the different approaches, a methodology similar to the one used for the filtering tests is proposed here, i.e., an evaluation of the results obtained by each aggregation of the S2 data (RMSE, r^2) against the d-GNSS measures.

III. Results

A. Impact of the NCC Parameters on the Estimated Displacement

1) Impact of the Search Distance: The search distance ($u_{\max}$ and $v_{\max}$) is an important parameter for NCC. If the surface displacement is larger than the search distance, areas of fast flow are not captured by the algorithm. Conversely, increasing the search distance significantly increases the computation time. Therefore, an unnecessarily large search distance wastes computing power and may limit the ability to compute large amounts of data.

Fig. 2 illustrates the flow velocity of glaciers in the Mont-Blanc massif considering two different search distances for the NCC algorithm (i.e., 4 × 4 and 32 × 32 pixels). It can be noted that large displacements are not captured with the small search window, because the glacier displacement at these locations is larger than the distance over which the algorithm computes the correlation function. Therefore, the search distance must be adjusted according to the local displacement. Since the glacier speed is not known a priori, and because surface displacements increase almost linearly with the temporal baselines between the images, the search distance is defined according to the temporal baseline in order to capture displacements at a rate of up to 1000 m per year, which is a maximum speed for alpine glaciers. Table III summarizes the search distance (in pixels) needed to capture the flow velocity of alpine glaciers with S2 (10-m resolution) depending on the temporal baseline between the used images and considering that the fastest surface flow velocity are 1000 m.yr^{-1}.

2) Impact of the Window Size: The $n_x \times n_y$ window size corresponds to the size of the image templates that are correlated. Large window size generally allows for better correlation of surface patterns and better estimate of the subpixel precision, but at expense of the spatial resolution of the output velocity fields. While resolution is less of an issue for large ice caps or ice sheets where glaciers are large relative to the sensor

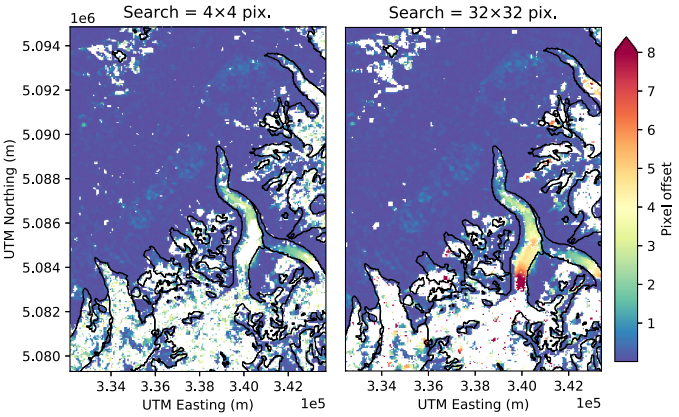


Fig. 2. Surface velocity of the Mont-Blanc massif considering different search distances applied to a 1-year image pair. Images were acquired in late summer (2018/08/28 and 2019/09/12). Search distances vary from 4 × 4 (left) to 32 × 32 (right), while window size and sampling resolution are kept constant with 16 × 16 and 5 × 5 pixels, respectively.

TABLE III

Search Distance Parameter Needed to Capture 1000 m.yr^{-1} With Sentinel-2. The Search Distance Is Round up to the Next Integer Value

Temporal baseline (days)	5	50	150	400
Search distance (pixel)	2	14	42	110

resolution, it can be a limiting factor for alpine glaciers that are less than a few hundred meters wide. Here, we test different $n_x \times n_y$ values for S2 image pairs acquired with about 1-year temporal baseline. Fig. 3 shows the glacier surface flow velocity obtained for the Mont-Blanc massif considering window sizes of 4 × 4, 8 × 8, 16 × 16, 32 × 32, 64 × 64 and 128 × 128 pixels. For 4 × 4 and 8 × 8 window sizes, the image correlation remains poor with less than 7.5% and 76% of correlated pixels, respectively, with SNR greater than 10. This limits the ability to distinguish correlated areas from those containing noise. For window size larger than 16 × 16, the percentage of correlated area is always higher than 85%. The largest windows (64 × 64 and 128 × 128) clearly smooth out the movement of the glaciers. For 16 × 16 and 32 × 32, results obtained seem relatively similar, but 16 × 16 provides a better output resolution than 32 × 32. Indeed, the wider the window used, the greater the possibility of having a subimage including both ice and nonice areas. The cross correlation algorithm will tend to converge toward a solution that corresponds to the ice-free parts because they usually contain patterns that are more pronounced and more stable over time than the ice-covered parts. Based on these results, the 16 × 16 pixel window size is to be preferred as it better preserves small structures while maintaining sufficient correlation. Nevertheless, it should be noted that 32 × 32 pixel window provides complementary results, especially for large glaciers.

3) Impact of the Sampling Resolution: The sampling resolution represents the step in the reference image R at which the NCC evaluation is performed. The NCC is generally not performed for each pixel of the R and S input images because

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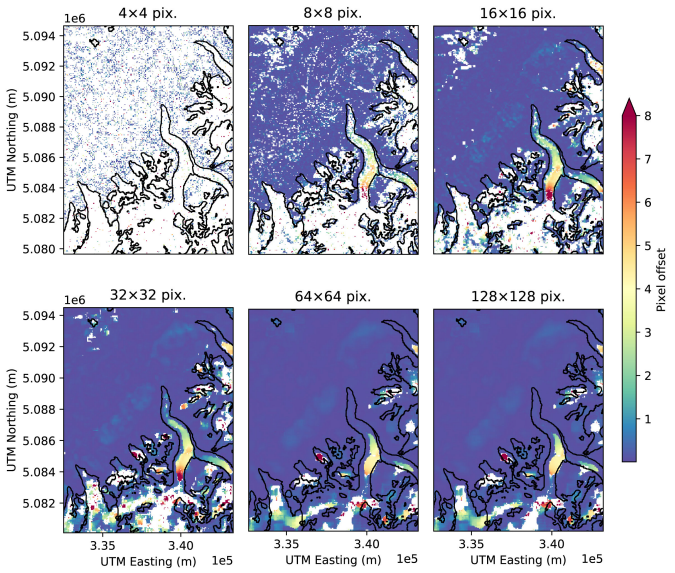


Fig. 3. Surface displacement within the Mont-Blanc massif considering different window sizes derived from a 1-year image pair. Images were acquired in late summer (2018/08/28 and 2019/09/12). Sampling resolution and search distances are kept constant for each test.

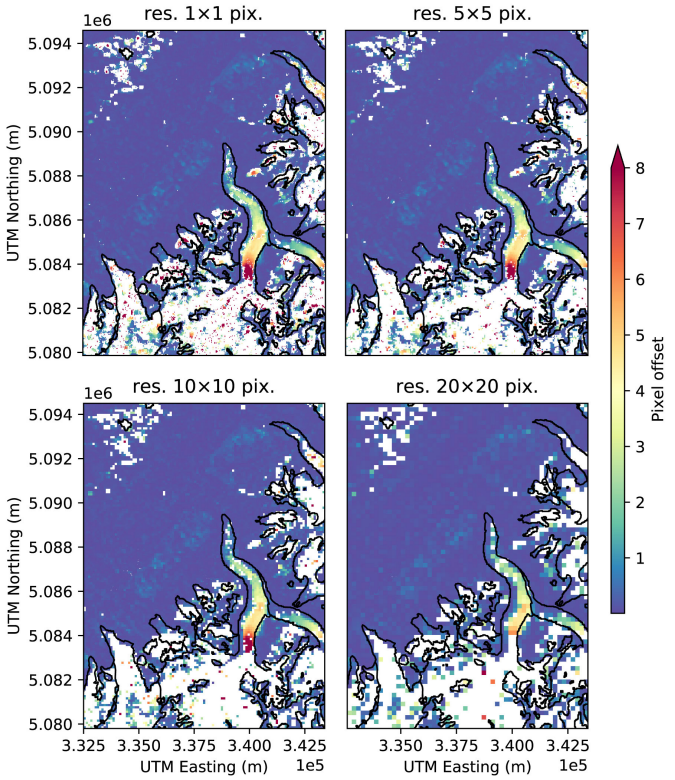


Fig. 4. Surface displacement considering different sampling resolutions. The test was made using the same images as in Section III-A2 using a pair with a 1-year temporal interval, and the optimal window size of 16 × 16 pixels as determined previously.

the computation would be too long and the results obtained from one pixel to another would be very redundant since the subimages that are correlated together remain largely the same. Here we test the influence of sampling resolution on the displacement field obtained from the same pair of S2 images. For the particular case of the Alps, this sampling resolution needs to be sufficiently small so that most glaciers are resolved. Therefore, the tested sampling resolutions are 1 × 1, 5 × 5, 10 × 10 and 20 × 20 pixels, corresponding to a spacing of 10 × 10, 50 × 50, 100 × 100 and 200 × 200 meters, respectively. For all tests, we use the optimal window size of 16 × 16 pixels as determined previously.

The glacier surface flow velocity maps in Fig. 4 show that 1 × 1 and 5 × 5 pixel sampling resolutions display the most realistic displacement field despite the fact the correlated subwindows (16 × 16) are largely redundant with their neighbors. Making a correlation at every image pixel takes about one hour for the small region of the Mont-Blanc area displayed in Fig. 4. Similar to the 1 × 1 results in terms of quality, the 5 × 5 takes only three minutes to compute (25 times less correlations to perform). For 10 × 10 sampling resolutions, some details of glacier surface displacements start to disappear and at 20 × 20 pixels the glaciers are difficult to resolve. Results from the 5 × 5 pixels sampling resolution clearly show the best compromise in terms of spatial coverage (number of points, n) vs. computational time.

4) Impact of the Temporal Baseline: In image correlation, the precision and accuracy at which surface displacements can be mapped depend directly on the resolution of the sensor and the time interval between the images to be correlated, i.e., temporal baseline or repeat cycle of the image pair. Indeed, it has been shown that matching algorithms can correlate surface patterns at subpixel precision σ_m of the order of 0.1–0.2 pixels [17], [23], [24]. Thus, for a temporal baseline of 30 days and a resolution of 10-m such as S2, that would represent an accuracy in speed σ of 12–24 m.yr^{-1}, assuming that $\sigma = \sigma_m.r/l$, where r is the sensor resolution and l the time lapse between two acquisitions. The two S2 satellites have, when combined, a nominal repeat cycle of five days. Image pairs with temporal baseline ranging from 5 days to 400 days are processed considering the optimized NCC parameters defined with the tests presented above, i.e., a search distance adapted to the temporal baseline, a 16 × 16 pixel window size and a 5 × 5 pixels sampling. In total, 18 126 image pairs are used over the period between 2015 and 2021 for this test.

For each temporal baseline, an average and standard deviation maps of the surface displacements are calculated. From the average maps, it appears that the mapping coverage varies significantly between short and long temporal baselines (Fig. 5). For the shortest temporal baselines (e.g., 15–25 days, 805 image pairs), the coverage is nearly complete but slow-moving areas are noisy. On the other hand, for the long temporal baselines around 1 year (360–370 days in Fig. 5, 625 image pairs) the fast-flowing areas are not captured, but when results are retrieved, their accuracy is improved. Intermediate temporal baselines (e.g., 165–175 days, 621 image pairs) result in cartographic coverage with large gaps, which is at least partly explained by changes in surface states (snow/ice) between seasons.

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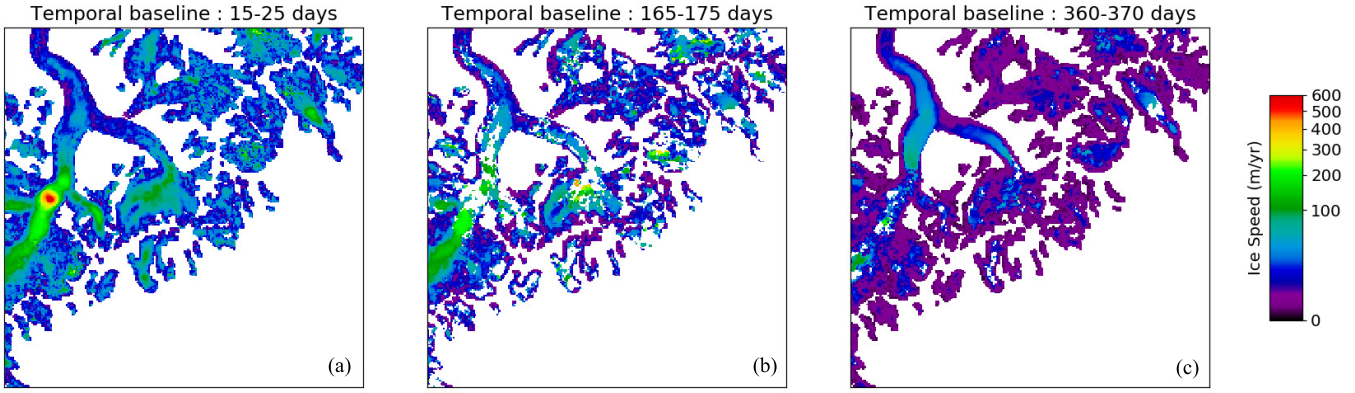


Fig. 5. Comparison between surface velocity maps over the Mer de Glace (Mont-Blanc massif, France) quantified from stacking of image pairs with 15-to-25 days interval (a), 165–175 days (b) and 360-to-370 days interval (c).

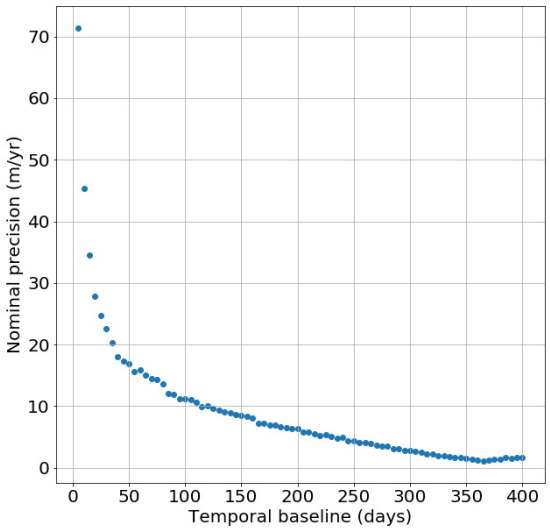


Fig. 6. Evolution of the precision of the ice velocity retrieved from Sentinel-2 for different temporal baselines (offset). Results obtained from the data retrieved from the Mont-Blanc test site. The nominal precision is defined as the average of the standard deviations between the multiple measurements obtained in ice-free areas (nonmoving).

From the standard deviation maps, we can estimate the precision of our mapping. Indeed, assuming that terrain around the glaciers should not move, the standard deviation obtained at these locations should reflect the precision. The evolution of the precision as a function of the length of the temporal baseline is shown in Fig. 6. From this graph, it can be seen that the precision for a temporal baseline of five days is about 80 m.yr^{-1} and then decreases exponentially to reach only a few meters per year for baselines longer than 300 days. The fact that the coverage and accuracy changes with the temporal baseline demonstrates that the data should be carefully combined for optimal results.

In summary, short temporal baselines (up to 25 days, Fig. 5) are suitable to capture fast velocities of several hundreds of m.yr^{-1}. Although rare on Alpine glaciers, such fast velocities can be found in ice falls, such as the upper reaches of Mer de Glace. For intermediate velocities ranging from 100 to 300 m.yr^{-1}, temporal baselines of up to 100 days seem suitable. For velocities less than 100 m.yr^{-1}, either temporal baselines between 100 and 150 days or between 250 and 300 days would be appropriate to provide precise and comprehensive measurements. Between 150 and 250 days, changes in the glacier surface characteristics (due to the seasonality) appear to limit the correlation (Fig. 5, central map). This analysis shows that the NCC on optical images can work for pairs of images separated by up to one year, due to the persistence of surface features over long periods for slow moving parts of the glaciers. Therefore, combining results from different temporal baselines will help to provide precise and comprehensive annually aggregated glacier surface flow velocity maps.

B. Data Reduction Results (Filtering and Aggregation)

Fig. 7 compares the annual glacier surface flow velocities at the level of ablation stakes located on the Mer de Glace and the Argentière Glacier (Fig. 1) quantified by d-GNSS measurements against those estimated from S2 (18 126 images pair in total) in this study. Fig. 7(a) shows the velocities extracted from S2 (before data reduction), and Fig. 7(b) shows the velocities finally retained after applying different filtering steps and aggregating the results into annual velocities considering the approach that provides the optimal results (OLS regression). Sections III-B1 and III-B2 present in detail the results obtained for each filtering and aggregation methods. It is worth noting that the pixels on the final glacier surface flow velocity maps are 50 × 50 m, while the measurements at the stakes are point measurements. This can of course introduce a bias in the comparison of the two measurements.

1) Results of the Filtering Approaches: Table IV summarizes the results obtained when filtering on the temporal baseline (see also Fig. S1). This filter becomes more selective as the parameter n increases. It appears that $n = 1$ shows the best performance, with RMSE and r^2 being improved compared to no filtering (i.e., the reference data called ref. in Table IV). Higher values of n decrease the filtering performances. It seems that for a too aggressive filtering ($n > 1$), the remaining S2 measurements become insufficient to accurately estimate the velocities measured by the d-GNSS.

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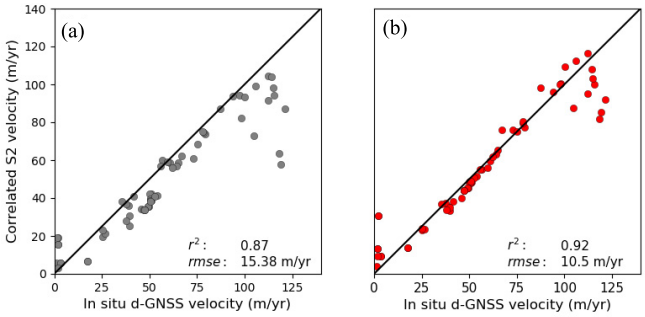


Fig. 7. Correlation between the level of annual glacier surface flow velocity quantified from in situ d-GNSS measurements and the measurements derived from S2 images before (a) and after (b) data filtering. The d-GNSS and S2 are measured at the ablation stakes located on the Mer de Glace and the Argentière Glacier.

TABLE IV

RESULTS OBTAINED FOR THE FILTERING ON THE TEMPORAL BASELINES FOR DIFFERENT VALUES OF THE FACTOR *n*

	ref.	n=1	n=2	n=3	n=4
RMSE (m/yr)	16.8	**14.2**	19.1	23.1	20.2
r^2	0.81	**0.86**	0.62	0.44	0.54
# of stakes	64	**64**	55	52	51

Table V summarizes the results in terms of RMSE and r^2 after filtering the S2-derived data for different ranges of angles around the median flow direction (see also Fig. S2). The angle value that gives the best results is ±45°, but statistics are almost similar for angle values ranging from ±25° to ±55°. Surprisingly, for the lowest angle range (±15° and ±5°), i.e., the most restrictive, the statistics are even worse than the reference. In these cases, the filtering becomes too selective, the few S2 measurements that are retained no longer provide good velocity estimates.

Table VI gives the statistical results for the tests on filtering the tails of the velocity distribution (see also Fig. S3). Removing 20% or 30% of both ends of the distribution (i.e., 10% or 15% at each ends, respectively) provides the best improvement in RMSE from 16.8 m.yr^{-1} without filtering to 12.3 and 12.2 m.yr^{-1}, respectively. The 20% filtering gives a slightly better r2 than the 30% filtering, so we select this first value for the application of this filter. With filtering greater than 30%, the RMSE and r^2 of the different tests degrade again. By filtering the velocity data too aggressively, we force the distributions to converge to the median values of the velocities obtained before filtering and consequently obtain worse RMSE and r^2 values than for lighter filtering such as 20% or 30%.

Finally, Table VII summarizes the statistical results (RMSE and r^2) when the three filters described above are combined (see also Fig. S4). All filter combinations show improved performance compared to the reference data. The best results are obtained when all filters are applied and for the case where the filter on temporal baseline and on distribution tails are combined. Indeed, in comparison with the in situ d-GNSS measurements, the satellite-derived velocity has an RMSE of 11.5 and 11.6 m.yr^{-1} with an r^2 of 0.9, respectively. Although the flow direction ("angle") filter shows improvements on the statistics when combined with the other filters, it is the filtering that provides the least benefit in terms of improvement of the results obtained. However, we prefer to keep it, because even if its interest appears to be limited in the case of this comparison with the d-GNSS measurements, it is likely to provide a significant improvement of the results on other noisy areas of the glaciers.

Based on the tests performed here, we decide that the most accurate results for the flow velocities derived from S2 images are obtained when a selection of the temporal baseline is performed, a filtering of the angles of plus or minus 45° around the median direction of the flow is performed and 20% of the extreme values of the distribution are removed.

2) Results of the Different Data Aggregation Approaches: Table VIII shows the results of the different aggregation methods in comparison to the d-GNSS measurements. For all these approaches, the same image processing and filtering steps on the obtained surface velocity have been applied (following the optimal parameters presented in Sections III-A1–III-A4). Although all RMSE values are very similar and the differences between them may not be significant, this analysis indicates that the annually averaged glacier surface flow velocities obtained applying the least-square linear regression provides the best agreement with the d-GNSS.

Fig. 8 shows a comparison between ice flow velocity maps obtained from a single pair of images acquired one year apart representing an annual velocity field and the corresponding annually aggregated glacier surface flow velocity maps obtained by applying different aggregation approaches. The map based on processing a single pair of images acquired on August, 28th 2018 and September, 12th 2019 shows incomplete mapping of flow velocities. The other maps are based on the results of all image pairs (4963 pairs in total) for the same period as the single pair of images. The different aggregation approaches show very similar results. The results are even closer when considering the two regression approaches (OLS and Theil-Sen) on the one hand and the two statistical approaches (median and weighted average) on the other hand (Fig. S5). Going into detail, slight differences can be noted on the slower parts of the glaciers and on the shear margins. In addition, the statistical approaches appear to be slightly noisier in the accumulation areas (see the red circle in Fig. 8 and the red/blue speckled areas in Fig. S5). Overall, the noisiest areas also correspond to parts of the glacier with a limited number of image pairs with a positive correlation (Fig. S6).

Fig. S6 also shows that the standard deviation map is spatially close to the ice velocity map, meaning that the standard deviation is higher in the fastest zones. However, when considering the coefficient of variation (also called relative standard deviation), the highest values (between 20% and 40%) are not found in the fastest areas but are more evenly distributed. This characteristic opens the way to the possible study of the intraannual variability of surface flow velocities.

IV. DISCUSSION

A. Implications for the Production of a Regional/Global Scale Product

We show here that with an optimized definition of the parameters of the correlation method used to derive these

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TABLE V

RESULTS OBTAINED FOR THE FILTERING CONSIDERING DIFFERENT RANGES OF ANGLES AROUND THE MEDIAN FLOW DIRECTION

	ref.	$\pm 65^o$	$\pm 55^o$	$\pm 45^o$	$\pm 35^o$	$\pm 25^o$	$\pm 15^o$	$\pm 5^o$
RMSE (m/yr)	16.8	16.7	15.5	**15.4**	15.4	15.4	17.9	31.3
r^2	0.81	0.78	0.81	**0.82**	0.81	0.80	0.73	0.53
# of stakes	64	64	64	**64**	64	64	64	63

TABLE VI

RESULTS OBTAINED FOR THE FILTERING ON THE TAILS OF THE DISTRIBUTIONS OF THE VELOCITY VALUES. THE PERCENTAGE INDICATED CORRESPONDS TO THE TOTAL AMOUNT OF FILTERED VALUES, I.E., HALF OF THIS VALUE FOR EACH END OF THE DISTRIBUTION

	ref.	5%	10%	15%	**20%**	30%	40%	50%
RMSE (m/yr)	16.8	15.6	15.9	14.3	**12.3**	12.2	13.4	14.8
r^2	0.81	0.83	0.82	0.86	**0.90**	0.89	0.90	0.85
# of stakes	64	64	64	64	**64**	64	64	63

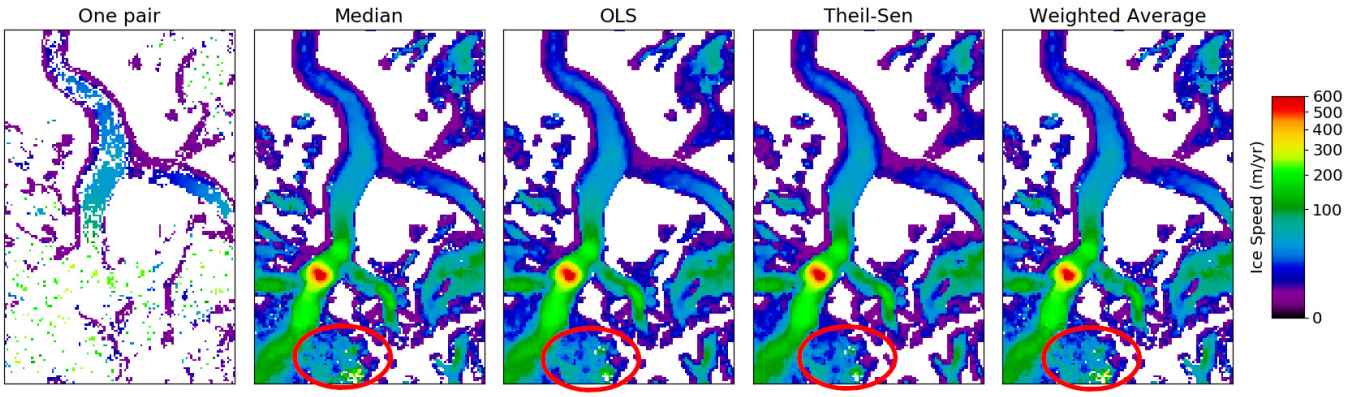


Fig. 8. Comparison between an ice velocity map obtained using images acquired one year apart on 2018/08/28 and 2019/09/12 (left) with maps issued from the different aggregation approaches applied on 4963 maps obtained during the same period (for Median and Weighted average methods), and interpolated from all dataset for OLS and Theil-Sen. The color coding is done on a logarithmic scale to highlight potential differences in slow moving parts.

TABLE VII

RESULTS OBTAINED WHEN COMBINING MULTIPLE FILTERS. "BASE," "TAILS," AND "ANGLE" REPRESENT FILTERS ON TEMPORAL BASELINE, THE DISTRIBUTION TAILS, AND THE FLOW DIRECTION, RESPECTIVELY

	ref.	**base. + angle**	**tails + angle**	**tails + base.**	**tails + base. + angle**
RMSE (m/yr)	16.8	13.2	13.2	11.5	**11.6**
r^2	0.81	0.87	0.87	0.90	**0.90**
# of stakes	64	64	64	64	**64**

TABLE VIII

RESULTS OBTAINED WHEN APPLYING DIFFERENT AGGREGATION APPROACHES ON THE FILTERED DATA

	Weighted average	**Theil-Sen regress.**	**Linear regress.**	**Median**
RMSE (m/yr)	16.7	11.7	**10.5**	11.6
r^2	0.78	0.90	**0.92**	0.90
# of stakes	64	64	**64**	64

velocity fields, it is possible to obtain velocity fields with an RMSE of about 10 m.yr^{-1} compared to the direct measurements made in the field. This approach tested here on the Mont-Blanc massif could be extended on a large scale to document the flow velocities of alpine massifs (such as the Alps, New Zealand, Andes, Caucasus, high mountains of Asia). Given the precision of the results obtained and the inhomogeneous temporal sampling of flow velocities for alpine glaciers, systematically obtained time series allowing to describe the intraannual evolution of glacier flow, as it can be done for the flow of ice caps, remains challenging. However, we show that statistical or regression approaches allow deriving fairly accurate annual measurements of the glacier surface flow velocity. It is noteworthy that the type of aggregation approach used to obtain these annual fields influences the results because of the assumptions made during this aggregation. Indeed, with linear least squares or Theil-Sen regressions, we assume that the evolution of velocities over time is linear. If these linear approaches allow exploiting all the measurements of a period and filling a period less rich in measurements through linear interpolation, it will prevent from revealing deviation of this linear trend (e.g., an acceleration of the flow for a given year). The low degree of freedom of the linear regression allows obtaining a robust final result despite relatively noisy data. One possibility to have less "rigid" regressions would be to use regressions with higher order functions (polynomial, spline, Gaussian process regression, e.g., [25], [26]) or to keep a linear approach but only applied locally, i.e., on a limited period or on the k-nearest neighbors (e.g., locally weighted scatterplot smoothing (LOWESS); [27]). The advantage of the latter nonparametric

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regressions would be to keep the simplicity and "rigidity" of linear regressions by having an estimation function which does not take a predetermined form, but is built according to the information coming from the data. In the case study presented here, the multiannual temporal evolution of the velocities according to the d-GNSS measurements seems to be quite close to linear variations during the period 2015 to 2021 justifying our hypothesis. Another possibility that we have tested (and which is finally quite close to local non-parametric regressions) is a simple statistical approach with a moving average and median [17], [22], [23]. Through these statistical indicators, no assumptions are made on the shape of the estimation function, but biases are more easily introduced by the inhomogeneous temporal sampling. For example, if the only available estimates of glacier surface flow velocities are for a limited period of time during the year, the annual average will reflect this and not represent the "real average flow" over the entire year. Regression methods will tend to take better account of these sampling biases. Finally, the fact that the results of the different aggregation methods are relatively similar after filtering suggests that strict filtering of the data has a greater influence than the aggregation methods themselves.

B. Potential Future Improvements

We show in this study that we are able to derive annual scale flow velocity fields of alpine glaciers. The main limitation of these observations, in terms of spatial and temporal resolutions of the time series obtained, comes from the resolution of the sensors used. Thus, with sensors twice the resolution (four times more information) of Sentinel-2, i.e., 5 m, it would be possible to follow more finely the small and/or slow flowing glaciers. This would also allow resolving more precisely intraannual variations, which remain for the moment difficult to document for this type of glacier. We can hope that future generations of satellites will go in this direction. Improvements could also be considered on the image processing. Here, we have considered the NCC between two images. This approach is well established and easily generalizable over large areas. In this study, the correlation is done between two windows (16×16 pixels), that is to say by exploiting the spatial dimension. For this, it is therefore necessary to have a relatively large spatial domain to obtain a significant correlation (see Fig. 3). Nevertheless, the acquisitions made by public platforms such as Landsat [National Aeronautics and Space Administration (NASA)] or Sentinel-2 (ESA) are now sufficiently recurrent to exploit the temporal dimension and so reduce the spatial one, allowing a possible gain in resolution in the velocity fields obtained. This possibility called ensemble matching or optical flow has already been successfully tested on specific targets by [11], [28] using optical images from Satellite pour l'observation de la Terre (SPOT)-5 and Sentinel-2, respectively. Altena and Kääb [28] demonstrate the possibility to generate a high-resolution (30 m) velocity field from high-repeat image sequences on a case study over the Khumbu ice fall (Everest region). Synthetic aperture radar (SAR), such as Sentinel-1, are another possibility to observe glacier surface flow velocities. SAR data have the advantage of not being affected by clouds and to be independent of sunlight. Sentinel-1 has a similar spatial resolution than Sentinel-2 or Landsat-8, with continuous acquisitions every 6 or 12 days. However, a drawback is that correlation requires spatially larger search windows than for optical data to obtain convincing results. This is not a major problem for the large glaciers in polar regions [23], [29], but is a strong limitation for the alpine glaciers [14]. One way to overcome this limitation for SAR sensors could be, in the same way as the work of [11], [28], to make ensemble matching allowing to reduce the spatial dimension in the correlator to a temporal dimension. This was proposed by [21] for a case study on Grosser Aletschgletscher (Swiss Alps). To improve the displacement estimates, they proposed a cross correlation stacking method that can significantly remove the background noise of the NCC. Instead of deriving the displacements of each NCC from single image pairs, they stack a series of consecutive pairwise NCCs and determine the displacement after averaging the stack. In this way, the tracking benefits from the redundant information of multiple NCCs and is more robust to noise. This could be a future way to better exploit SAR data to complement the results obtained by optical sensors on these alpine glaciers.

V. Conclusion

With the current resolution of public satellites constellations (e.g., Landsat or Sentinel), mapping mountain glacier surface flow velocity remains challenging by image correlation. Indeed, the characteristic size of these objects is of the same order of that of the windows used for cross-normalized correlation. In this study, we first evaluated the most suitable parameters to correlate Sentinel-2 images on alpine glaciers such as those of the Mont-Blanc massif; and then we tested different methods to derive annual glacier surface flow velocity maps using the large number of acquisitions. In the phase of evaluating the parameters of the S2 image correlation, we found that the most suitable parameters for our case were the correlation of search windows of 160×160 meters every 50 m with a search distance adapted to the displacement of the studied glaciers. Using these parameters, we calculated a large number of velocity fields between 2015 and 2021 with temporal baselines ranging from 5 to 400 days. We show that these data are noisy and require strict filtering combining selection on flow directions, on temporal baselines and on extreme values to obtain the most accurate results. After this filtering, tests on different aggregation methods in order to obtain annual values show only small differences with a slightly better result for the OLS regression. This indicates that accurate reconstruction of velocity fields should involve strict data filtering rather than optimization of aggregation methods.

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